

MIT651 — Capstone Project 1

Group Assignment 1 — Project Proposal

Stock Market Prediction Using Machine Learning and Probabilistic Analysis

Group Name: [Group 02]

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Name	Student ID	Email	Role
Chandra Laxmi napit	S20250075	S20250075@sihe.edu.au	Team Leader
Sayed md asif same	S20250066	S20250066@sihe.edu.au	Team Member
Mushfikul Islam	S20250051	S20250051@sihe.edu.au	Team Member

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A. Background

Project: Stock Market Prediction Using Machine Learning and Probabilistic Analysis

A.1 Problem Identification

Stock prices and market indices change continuously in response to company information, investor expectations, macroeconomic conditions, liquidity, market sentiment and unexpected events. This makes financial time series noisy, non-stationary and difficult to predict reliably. Many simple prediction tools provide only a single estimated future value or direction. A single output can be misleading because it does not show how uncertain the forecast is. The proposed project addresses this limitation by combining machine-learning based prediction with probabilistic analysis so that the system can present both an expected forecast and a clear measure of uncertainty.

The planned system is not intended to provide guaranteed investment advice. Its purpose is to support analytical study by allowing users to select a stock, process historical market data, run one or more machine-learning models, and inspect a probability, prediction interval or related uncertainty measure together with model evaluation metrics. This creates a more informative decision-support workflow than a point forecast alone.

A.2 Preliminaries of the Problem Domain

A stock-market dataset commonly contains daily open, high, low, close and volume (OHLCV) values. These raw values can be transformed into returns, moving averages, volatility measures, momentum indicators and lagged variables. Because market observations are ordered in time, the training and testing process must preserve chronological order. Randomly mixing future observations into the training set can produce data leakage and an unrealistic estimate of model performance.

Machine-learning methods can model nonlinear relationships between engineered features and a future target such as next-day return, direction or closing price. Tree-based methods provide a useful baseline for structured features, while recurrent networks such as long short-term memory (LSTM) models are designed to learn temporal dependencies. However, financial forecasts remain uncertain even when point-prediction accuracy is acceptable. Probabilistic analysis therefore adds a second layer of information by estimating the distribution or range of plausible outcomes. Examples include class probabilities, Bayesian predictive distributions, Monte Carlo dropout, quantile models or conformal prediction intervals.

A.3 Existing Related Work and Drawbacks

Research in financial forecasting shows continued use of LSTM, tree-based models, hybrid deep-learning approaches and feature-engineering methods. The literature also highlights major problems: non-stationarity, overfitting, changing market regimes, data leakage, transaction costs and weak out-of-sample generalisation. A further limitation is that many studies focus mainly on point accuracy and give less attention to uncertainty calibration, even though uncertainty is important when a forecast is used for risk-sensitive decisions.

Related work	Approach	Relevance	Main drawback
Fischer and Krauss (2018)	LSTM	Large-scale directional prediction on S&P 500 constituents.	Performance changed over time; profitable signals weakened in later periods and transaction costs matter.
Sezer et al. (2020/2023 survey context)	Deep-learning review	Summarises CNN, LSTM and hybrid approaches for financial prediction.	Reported performance can be difficult to compare because datasets, validation procedures and metrics vary.
Chandra and He (2021)	Bayesian neural network	Provides stock forecasts with explicit predictive uncertainty.	Bayesian inference is more computationally involved and interval quality depends on model assumptions.

Htun, Biehl and Petkov (2023)	Feature selection/extraction survey	Shows the importance of selecting informative market features.	Feature usefulness can change across assets and market regimes.
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Table 1. Selected related work and identified drawbacks

The key gap motivating this project is therefore not the claim that machine learning can perfectly predict the stock market. Instead, the project focuses on a transparent workflow that combines prediction, time-aware evaluation and uncertainty information. This is more realistic for a capstone prototype because the system can demonstrate both what the model predicts and how confident or uncertain that prediction appears to be.

A.4 Proposed Approach

The proposed project will develop a web-based stock market prediction prototype. Historical market data will be collected for a selected symbol and date range, validated and transformed into model-ready features. At least one baseline machine-learning model and one sequence-aware model will be evaluated using time-based train/validation/test splits. The better-performing workflow will then be connected to a probabilistic component that produces a confidence estimate or prediction interval. The user interface will present the forecast, uncertainty, evaluation metrics and historical forecast records. As this document is a proposal, no final prediction accuracy or investment return is claimed at this stage.

B. User Stories and Sprints

This section presents the planned user stories and sprint structure for the proposed Stock Market Prediction Using Machine Learning and Probabilistic Analysis system. The user stories define the expected functionalities of the system and are organized using Jira as the project management platform. The product backlog is divided into five planned sprints to support incremental development and continuous evaluation.

B.1 Project Objectives

Based on the proposed system scope, the following objectives are defined:

OBJ1: To allow users to select a stock and retrieve historical market information.

OBJ2: To preprocess stock data and generate meaningful technical features.

OBJ3: To develop a machine-learning workflow for predicting future stock movement.

OBJ4: To provide probability-based prediction analysis along with the predicted direction.

OBJ5: To present prediction results through a clear dashboard containing charts and indicators.

OBJ6: To allow users to review previous prediction records and details.

OBJ7: To provide administrative functionality for managing stocks, models, and stored records.

B.2 Product Backlog

The product backlog contains the planned system functionalities expressed as user stories. Each story represents a specific requirement from the perspective of a user or administrator.

Jira Key	User Story	Planned Sprint
SMP-6	As a user, I want to select a stock so that I can request a prediction for a specific company.	Sprint 1
SMP-7	As a user, I want the system to retrieve historical stock data so that prediction can be based on previous market behaviour.	Sprint 1
SMP-8	As a user, I want the system to validate market data so that incomplete information is not used.	Sprint 1
SMP-9	As a user, I want the system to generate market features so that the ML model can analyse stock patterns.	Sprint 2
SMP-10	As a user, I want the system to apply a trained ML model so that future stock direction can be estimated.	Sprint 2
SMP-11	As a user, I want to view predicted direction so that I can understand the forecast.	Sprint 3

SMP-12	As a user, I want to view prediction probability so that I can understand uncertainty.	Sprint 3
SMP-13	As a user, I want to view charts and indicators so that I can interpret the prediction.	Sprint 3
SMP-14	As a user, I want to view previous prediction results so that I can review earlier analysis.	Sprint 4
SMP-15	As a user, I want to view prediction details so that I can analyse previous outputs.	Sprint 4
SMP-16	As an administrator, I want to manage stocks and models so that the system can be maintained.	Sprint 5
SMP-17	As an administrator, I want to manage prediction records so that system information remains organized.	Sprint 5

B.3 Jira Product Backlog Evidence

All product backlog stories are created and managed in Jira. Screenshots of the Jira backlog will be included below.

The screenshot shows the Jira Product Backlog for a project named 'Stock Market Prediction'. The interface includes a top navigation bar with tabs for Summary, List, Board, Backlog (selected), Calendar, Timeline, Docs, Forms, and Development. Below the navigation bar is a search bar and a filter button. The backlog list contains several items, each with a status icon, a title, a progress bar, a priority dropdown, and a 'To Do' button. The items are:

- SMP-6 Select Stock (Status: In Progress, Progress: 50%, Priority: High, To Do)
- SMP-7 Retrieve Historical Stock Data (Status: To Do, Progress: 0%, Priority: High, To Do)
- SMP-8 Validate Market Data (Status: In Progress, Progress: 50%, Priority: High, To Do)
- SMP-9 Generate Market Features (Status: In Progress, Progress: 50%, Priority: High, To Do)
- SMP-10 Run Machine Learning Prediction (Status: In Progress, Progress: 50%, Priority: High, To Do)
- SMP-11 View Predicted Direction (Status: To Do, Progress: 0%, Priority: High, To Do)
- SMP-12 View Prediction Probability (Status: To Do, Progress: 0%, Priority: High, To Do)
- SMP-13 View Charts and Indicators (Status: To Do, Progress: 0%, Priority: High, To Do)

Figure B1. Jira Product Backlog – User Stories SMP-6 to SMP-13

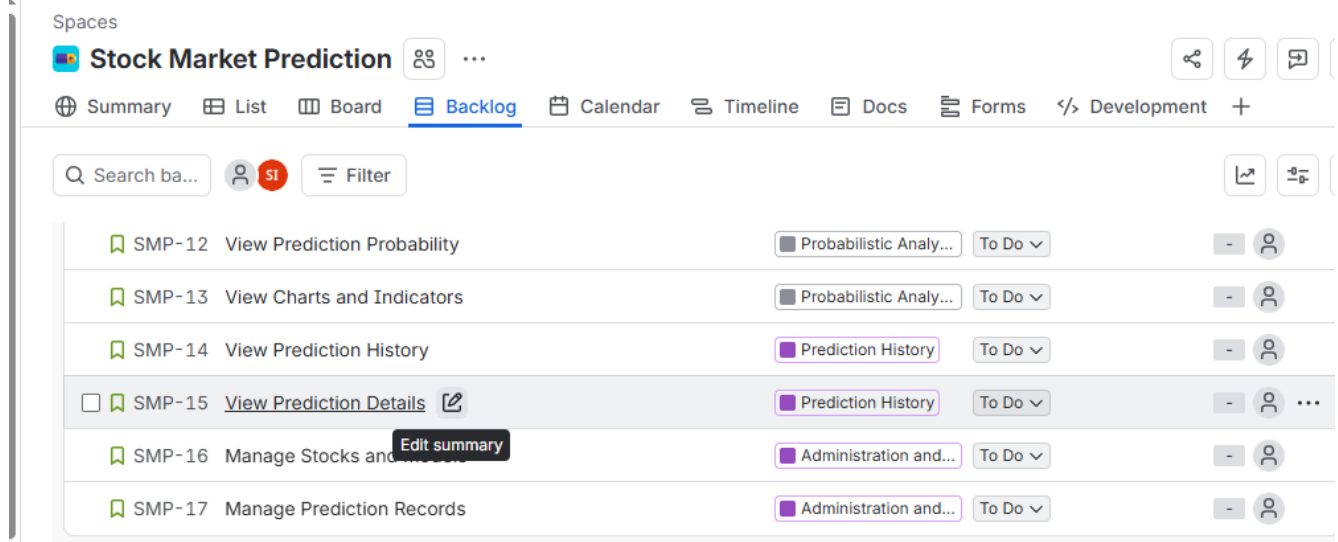


Figure B2. Jira Product Backlog – User Stories SMP-12 to SMP-17

B.4 Five-Sprint Schedule

The project is planned across five sprints covering the complete development cycle of the Stock Market Prediction Using Machine Learning and Probabilistic Analysis system. The schedule begins from the initial system design and data preparation phase and continues through model development, probabilistic analysis, interface development and final integration. Each sprint is designed around a specific functional milestone and produces a demonstrable system increment.

Sprint	Course / Demo	Start Date	End Date	User Stories	Planned Deliverable
Sprint 1	MIT651 – Week 8 Demo	17 Jul 2026	11 Sep 2026	SMP-1, SMP-2, SMP-3	Stock selection interface, historical market-data retrieval, data validation and error handling
Sprint 2	MIT651 – Week 12 Demo	12 Sep 2026	9 Oct 2026	SMP-4, SMP-5, SMP-6	Data preprocessing, feature engineering, machine-learning prediction workflow and model comparison
Sprint 3	MIT652 – Week 4 Demo	16 Nov 2026	11 Dec 2026	SMP-7, SMP-8, SMP-9	Probabilistic prediction analysis, evaluation metrics and prediction result interface
Sprint 4	MIT652 – Week 8 Demo	12 Dec 2026	8 Jan 2027	SMP-10, SMP-11	Prediction history management and predicted-versus-actual market performance comparison
Sprint 5	MIT652 – Week 12 Demo	9 Jan 2027	5 Feb 2027	SMP-12, SMP-13	Administration, final system integration, testing and final

					demonstration preparation
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Table B2. Planned Five-Sprint Schedule

B.5 Initial Five Sprint Backlogs

The initial sprint backlogs distribute the product backlog according to functional dependency and the required demonstration milestones. Each sprint is designed to produce a usable system increment rather than a collection of disconnected technical tasks. The development begins with data acquisition and gradually progresses towards prediction, uncertainty analysis, history management and final system integration.

B.5.1 Sprint 1 – Market Selection and Data Validation

Jira Key	Sprint Backlog Item
SMP-6	Stock selection and prediction request
SMP-7	Historical stock-market data retrieval
SMP-8	Data validation and invalid-data feedback

Planned period: 17 July 2026 to 11 September 2026.

Planned sprint outcome: At the end of Sprint 1, users should be able to select a target stock, specify the required prediction request and retrieve validated historical market data. The system should also provide appropriate feedback when the collected data is incomplete or unsuitable for analysis.

B.5.2 Sprint 2 – Machine Learning Prediction Workflow

Jira Key	Sprint Backlog Item
SMP-9	Data preprocessing and feature engineering
SMP-10	Machine-learning based stock prediction
SMP-11	Prediction model comparison and evaluation

Planned period: 12 September 2026 to 9 October 2026.

Planned sprint outcome: At the end of Sprint 2, the core prediction workflow should be available. The system should preprocess historical market data, generate relevant features, apply machine-learning models and compare model performance using appropriate evaluation methods.

B.5.3 Sprint 3 – Probabilistic Forecast Presentation

Jira Key	Sprint Backlog Item
SMP-12	Prediction uncertainty and probability estimation
SMP-13	Accuracy evaluation and calibration measures
SMP-14	Forecast result and uncertainty interface

Planned period: 16 November 2026 to 11 December 2026.

Planned sprint outcome: At the end of Sprint 3, the system should present the predicted stock movement together with probability-based analysis. Users should be able to understand prediction confidence, evaluation results and supporting visualization.

B.5.4 Sprint 4 – Forecast History and Review

Jira Key	Sprint Backlog Item
SMP-15	Saved prediction history
SMP-16	Predicted-versus-realised market comparison

Planned period: 12 December 2026 to 8 January 2027.

Planned sprint outcome: At the end of Sprint 4, users should be able to review previous prediction records and compare predicted stock movement with actual market outcomes.

B.5.5 Sprint 5 – Administration and Final Integration

Jira Key	Sprint Backlog Item
SMP-17	User, stock and dataset administration
SMP-18	Model configuration and prediction-record administration

Planned period: 9 January 2027 to 5 February 2027.

Planned sprint outcome: At the end of Sprint 5, administrative features, final system integration, testing and demonstration preparation should be completed.

B.6 First Sprint Scope and Jira Evidence

The first sprint is planned from 17 July 2026 to 11 September 2026 and contains SMP-1, SMP-2 and SMP-3. These stories establish the initial user interaction required before stock prediction can occur: selecting a target stock, retrieving historical market data and validating the collected information. This provides a coherent first increment for the MIT651 Week 8 demonstration.

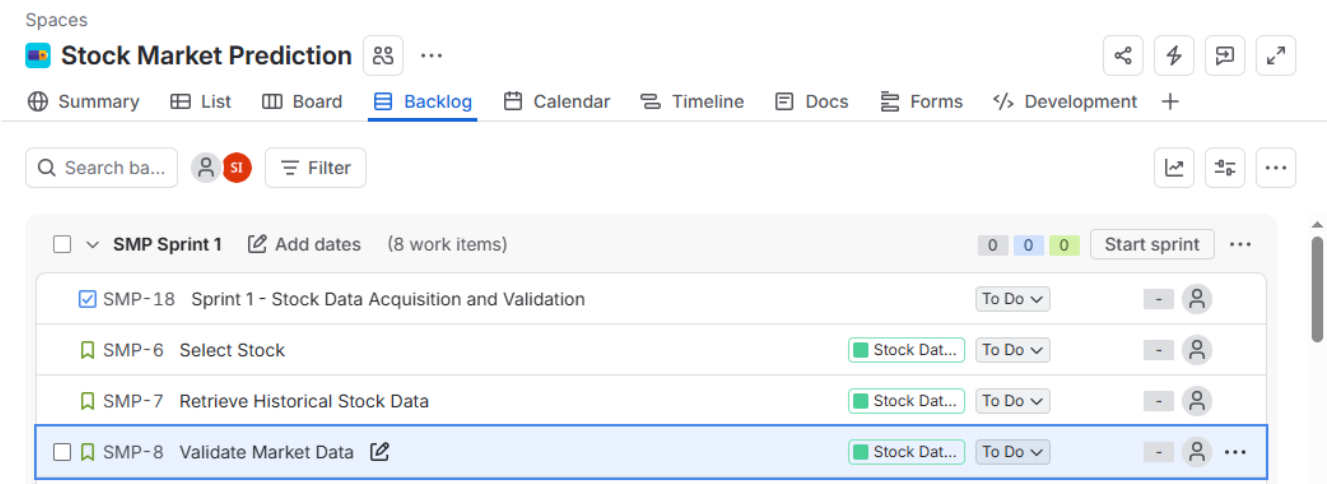


Figure B4. User Stories Allocated to Sprint 1 in Jira

B.7 Project Objective–User Story Mapping

Table B3 demonstrates how the product backlog covers the defined project objectives. This mapping ensures that each objective is represented by one or more planned user-facing functions.

Objective	Objective Summary	Supporting User Stories
OBJ1	Enable users to select stocks and retrieve historical market data	SMP-6, SMP-7
OBJ2	Develop the machine-learning based stock prediction workflow	SMP-9, SMP-10
OBJ3	Validate and prepare market data before prediction	SMP-8, SMP-9
OBJ4	Provide stock movement prediction results	SMP-11

OBJ5	Provide probability analysis and supporting visualization	SMP-12, SMP-13
OBJ6	Allow review of previous prediction records	SMP-14, SMP-15
OBJ7	Provide appropriate administrative functionality	SMP-16, SMP-17

Table B3. Mapping of Project Objectives to User Stories

B.8 Planned Novel Functionality Compared with Existing Systems

Novelty is considered relative to the limitations of existing stock prediction systems identified in the background section. The main planned differentiator is an integrated prediction workflow that combines historical market-data analysis, machine-learning prediction and probability-based interpretation within a single user-facing system.

User Story	Planned Differentiating Functionality	Explanation
SMP-10	Machine-learning based prediction workflow	The proposed system is intended to combine feature-based market analysis with machine-learning models .The machine learning models generate future stock movement predictions rather than relying only on traditional statistical indicators.
SMP-11	Probability-based prediction interpretation	The prediction workflow is planned to provide probability values along with predicted direction. This allows users to understand the confidence and uncertainty associated with the generated forecast.
SMP-9 and SMP-10	Integrated prediction-history review	The proposed application plans to retain and present previous prediction records within the same user-facing system. This allows the users to review earlier forecasts and analyse previous prediction performance.

Table B4. Planned Novel or Differentiating Functionality

C. Project Planning

This section presents the initial planning for the proposed project. The Work Breakdown Structure divides the work into manageable packages. The Gantt chart shows the planned sequence and duration of the main activities. The plan follows the requested order like requirements are placed under Planning, System Design occurs before Development, Prototype occurs before Testing, no separate Results task is included and Documentation begins from the start of the project.

C.1 Work Breakdown Structure (WBS)

The WBS organises the project into seven major areas include Planning, Data Preparation, System Design, Development, Prototype, Testing & Evaluation and Documentation. Documentation is not treated as a final-only activity. It begins with the proposal and continues throughout the project lifecycle.

1. Planning	Topic selection, requirements analysis, project objectives, background and related work, user stories,sprint planning
2. Data Preparation	Market-data acquisition, cleaning, missing-value handling, return calculation, technical feature engineering
3. System Design	Three-layer architecture, data flow, database design, UI and storyboard design, model and probabilistic workflow design
4. Development	Data pipeline; baseline ML model; LSTM/sequence model; probabilistic analysis module; API/backend; forecast interface.
5. Prototype	Initial prototype, component integration, user-flow refinement and prototype review
6. Testing & Evaluation	Functional testin, time-series backtesting, prediction metrics, probabilistic calibration, integration/usability testing and issue resolution
7. Documentation	Initial proposal documentation, ongoing technical documentation, model/data notes, final documentation and review

Figure C1. Work Breakdown Structure of the Proposed Project

C.2 Gantt Chart

The Gantt chart presents the initial sequence of project activities. Planning begins on 17 July 2026, followed by data preparation, system design, development, prototype development, and testing and evaluation. Documentation starts from the beginning of the project and continues in parallel with the other activities. The schedule intentionally places system design before development and prototype development before formal testing. The separate “Results” task has been removed because this is a project proposal. The schedule may be adjusted to remain consistent with the five sprint demonstrations required in MIT651 and MIT652.

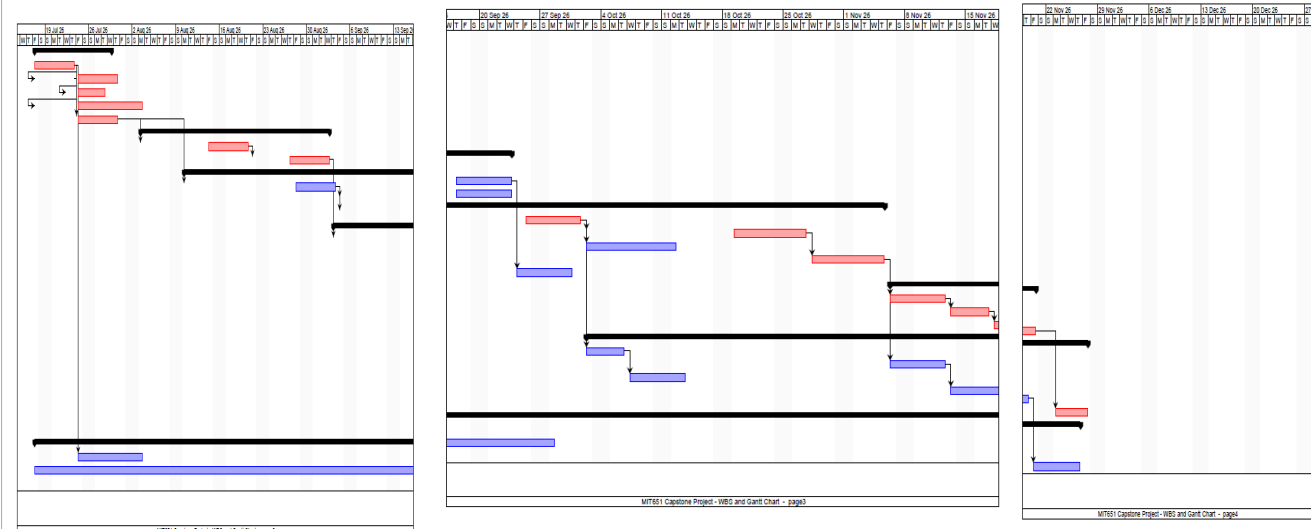


Figure C1: Gantt chart

		Name	Duration	Start	Finish	Predecessors	Resource Names
1		Planning	9 days?	7/17/26, 8:00 AM	7/29/26, 5:00 PM		
2		Topic Selection	5 days?	7/17/26, 8:00 AM	7/23/26, 5:00 PM		
3		Requirements Analysis	5 days?	7/17/26, 8:00 AM	7/23/26, 5:00 PM	2	
4		Define Project Objectives	3 days?	7/22/26, 8:00 AM	7/24/26, 5:00 PM	3	
5		Background and Related...	7 days?	7/17/26, 8:00 AM	7/27/26, 5:00 PM	2	
6		User Stories and Sprint P...	5 days?	7/23/26, 8:00 AM	7/29/26, 5:00 PM	3	
7		Data Preparation	23 days?	8/3/26, 8:00 AM	9/2/26, 5:00 PM		
8		Data Collection / Acquisit...	5 days?	8/3/26, 8:00 AM	8/20/26, 5:00 PM	6	
9		Data Preparation / Proce...	5 days?	8/21/26, 8:00 AM	9/2/26, 5:00 PM	8	
10		System Design	33 days?	8/10/26, 8:00 AM	9/23/26, 5:00 PM		
11		System Architecture Des...	5 days?	8/10/26, 8:00 AM	9/3/26, 5:00 PM	6	
12		UI and Storyboard Design	5 days?	9/4/26, 8:00 AM	9/23/26, 5:00 PM	11	
13		Classification Workflow ...	5 days?	9/4/26, 8:00 AM	9/23/26, 5:00 PM	11	
14		Development	46 days?	9/3/26, 8:00 AM	11/5/26, 5:00 PM		
15		MRI Input and Validation...	5 days?	9/3/26, 8:00 AM	10/1/26, 5:00 PM	9	
16		Spatial Component Deve...	7 days?	10/2/26, 8:00 AM	10/27/26, 5:00 PM	15	
17		Sequential Transformer ...	7 days?	10/2/26, 8:00 AM	10/12/26, 5:00 PM	15	
18		Hybrid Model Integration	7 days?	10/28/26, 8:00 AM	11/5/26, 5:00 PM	16	
19		Result Interface Develo...	5 days?	9/24/26, 8:00 AM	9/30/26, 5:00 PM	12	
20		Prototype Development	11 days?	11/6/26, 8:00 AM	11/20/26, 5:00 PM		
21		Build Initial Prototype	5 days?	11/6/26, 8:00 AM	11/12/26, 5:00 PM	18	
22		Prototype Integration	3 days?	11/13/26, 8:00 AM	11/17/26, 5:00 PM	21	
23		Prototype Review and R...	3 days?	11/18/26, 8:00 AM	11/20/26, 5:00 PM	22	
24		Testing and Evaluation	41 days?	10/2/26, 8:00 AM	11/27/26, 5:00 PM		
25		Test Input and Validatio...	3 days?	10/2/26, 8:00 AM	10/6/26, 5:00 PM	15	
26		Test Classification Workf...	5 days?	11/6/26, 8:00 AM	11/12/26, 5:00 PM	18	
27		System Integration Testing	5 days?	10/7/26, 8:00 AM	10/13/26, 5:00 PM	25	
28		Performance Evaluation	5 days?	11/13/26, 8:00 AM	11/19/26, 5:00 PM	26	
29		Resolve Identified Issues	5 days?	11/23/26, 8:00 AM	11/27/26, 5:00 PM	23	
30		Documentation	95 days?	7/17/26, 8:00 AM	11/26/26, 5:00 PM		
31		Initial Proposal Documen...	7 days?	7/24/26, 8:00 AM	8/3/26, 5:00 PM	2	
32		Ongoing Technical Docu...	52 days?	7/17/26, 8:00 AM	9/28/26, 5:00 PM		
33		Final Documentation and...	5 days?	11/20/26, 8:00 AM	11/26/26, 5:00 PM	28	

MIT651 Capstone Project - WBS and Gantt Chart - page 1

D. System Architecture

The proposed system uses a three-layer architecture separating user facing presentation functions, analytical/business logic and stored data. This separation is intended to make the capstone prototype easier to develop, test and extend across the five planned sprints.

D.1 Overall System Architecture

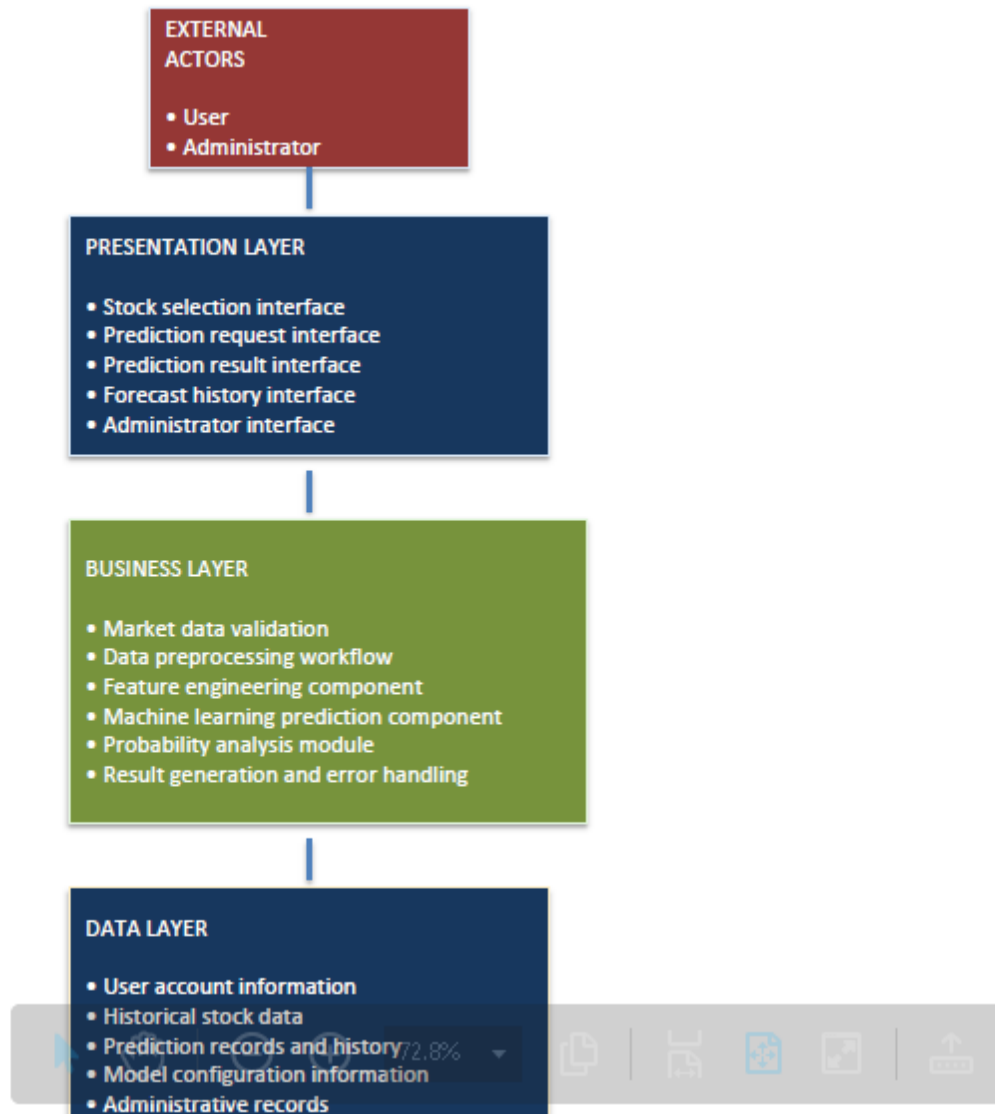


Figure D1. Proposed three-layer system architecture

D.2 Presentation Layer

The presentation layer provides the web interfaces through which users and administrators interact with the system. It collects user input and displays outputs without containing the main prediction logic.

- Stock selection and date-range interface allows a user to select a market symbol, historical period and forecast horizon.
- Data status interface reports whether historical data were retrieved and validated successfully.
- Prediction interface allows the user to select or run an available model.

- Forecast and uncertainty interface presents the predicted value or direction together with probability or interval information.
- History and comparison interface allows users to review saved forecasts and compare them with realised values.
- Administrator interface supports authorised management of users, datasets, model versions and forecast records.

D.3 Business / Analytics Layer

The business and analytics layer contains the main application logic and coordinates the prediction workflow. It validates requests, prepares market data, constructs features, invokes the chosen model, calculates probabilistic uncertainty, evaluates model performance and communicates with the data layer when information must be stored or retrieved.

- Data validation include checks symbol availability, chronological order, missing values and minimum history requirements.
- Preprocessing and feature engineering include produces returns, lagged variables, moving averages, volatility and other selected indicators.
- Machine-learning prediction engine include executes baseline and sequence-aware models according to the selected configuration.
- Probabilistic analysis module include produces class probabilities, predictive intervals or another suitable uncertainty representation.
- Evaluation module include computes time-aware test metrics and probability calibration measures.
- Application services and error handling include return outputs to the user interface and generate meaningful failure messages.

D.4 Data Layer

The data layer manages the information required by the application. The final schema will be refined during System Design, but the proposed database will store application records rather than performing the prediction itself.

- User account and access information.
- Market symbol and dataset metadata.
- References to cached or prepared historical data.
- Model version and configuration records.
- Generated forecast, probability/interval and evaluation records.
- Administrative and audit information.

D.5 External Actors and Their Interaction with the System

External Actor	Main Interactions	Related User Stories
Investor / Analyst	Selects a stock and date range, starts data processing and prediction, views forecast and uncertainty, compares model performance and reviews previous forecasts.	SMP-1 to SMP-11
Administrator	Manages authorised users, stored datasets, model versions and forecast records.	SMP-12 to SMP-13
Market Data Provider	Provides historical market data requested by the backend. It is an external service rather than an internal system component.	Supports SMP-2

D.6 Use Case Diagrams

Two use case diagrams are included. The first shows the overall interaction between the main user types and the proposed system. The second expands the analytical workflow to show the major steps involved in generating a risk-aware stock prediction.

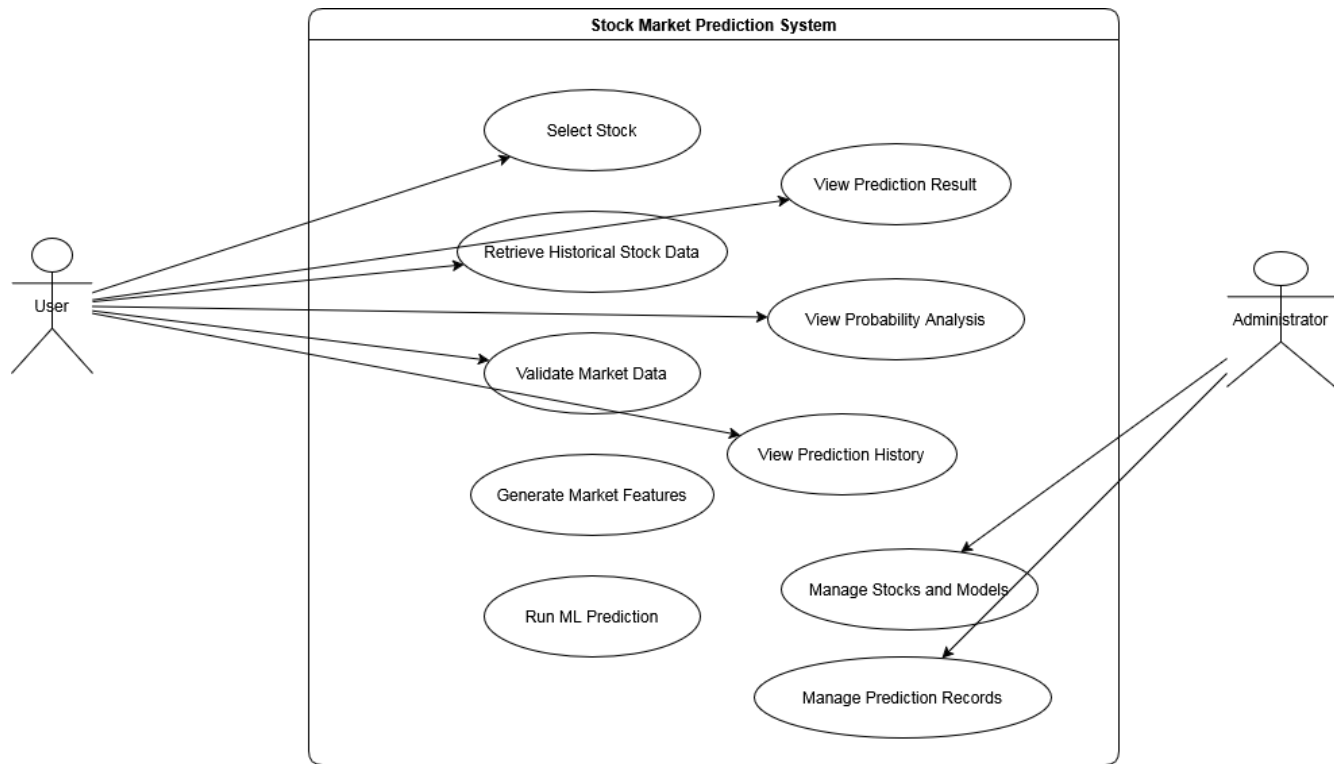


Figure D2. Overall use case diagram for the proposed system

D.7 Planned Technologies and Programming Languages

System Area	Planned Technology / Language	Planned Purpose
Core programming and analytics	Python	Main language for data processing, machine learning, evaluation and probabilistic analysis.
Machine learning	scikit-learn	Baseline regression/classification models, preprocessing and evaluation utilities.
Deep learning	PyTorch	Implementation of an LSTM or other sequence-aware model where required.
Data processing	pandas and NumPy	Time-series manipulation, feature engineering and numerical operations.
Probabilistic analysis	PyTorch distributions / quantile or conformal library as selected	Prediction probabilities or intervals and uncertainty-oriented evaluation.
Backend / API	FastAPI	Application endpoints connecting the presentation layer with Python analytics.
Frontend	React with HTML/CSS/JavaScript	Web dashboard for market selection, forecast visualisation, history and administration.
Database	PostgreSQL	Storage of users, model metadata, forecast history and administrative records.
Version control	Git and GitHub	Source control and team collaboration.
Agile project management	Jira	Backlog, user stories, sprint allocation and progress tracking.
Project planning	ProjectLibre	Work breakdown and Gantt planning for the project schedule.

D.8 Third-Party Functionality

Third-Party Component	Type	Planned Use
pandas / NumPy	Open-source Python libraries	Market-data preparation, feature engineering and numerical processing.
scikit-learn	Open-source ML library	Baseline models, scaling, evaluation and time-series validation utilities.
PyTorch	Open-source deep-learning framework	Sequence model development and probabilistic tensor operations.
yfinance or equivalent market-data provider	External data library/service	Historical OHLCV retrieval for selected market symbols; final provider is subject to availability and terms.
FastAPI	Open-source web framework	Backend API endpoints.
React	Open-source JavaScript library	Reusable user-interface components.
PostgreSQL	Open-source database	Persistent storage for application records.
GitHub	External development service	Repository hosting and collaboration.
Jira	External project-management service	Product backlog, sprint planning and progress tracking.
ProjectLibre	Desktop project-planning software	Gantt and schedule planning.

D.9 Planned Technology Interaction

A request begins in the React presentation layer and is sent to the FastAPI backend. Python services retrieve and validate market data, pandas/NumPy build the model inputs, scikit-learn or PyTorch executes the selected model, and the probabilistic component estimates uncertainty. Forecast records and model metadata are stored in PostgreSQL. The user then receives the prediction, uncertainty information and selected performance metrics through the web interface.

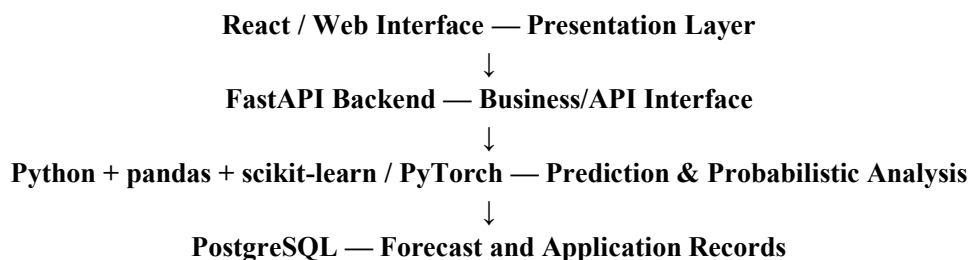


Figure D4. Planned interaction between the main technologies

E. User Interface and Flow Diagrams

This section presents the proposed user interface storyboards and interaction flows for the system. The storyboards illustrate how users and administrators will interact with the application to upload MRI scans, initiate classification, view generated results, review previous records and perform administrative activities. Multiple user stories are grouped within each storyboard where appropriate so that all user stories in the product backlog are covered.

E.1 Storyboard 1 - MRI Upload and Validation

This storyboard illustrates how a user uploads a cross-sectional 3D MRI scan and how the system validates the selected file before allowing the classification process to continue. If the file is invalid, the user is informed and can select another file.

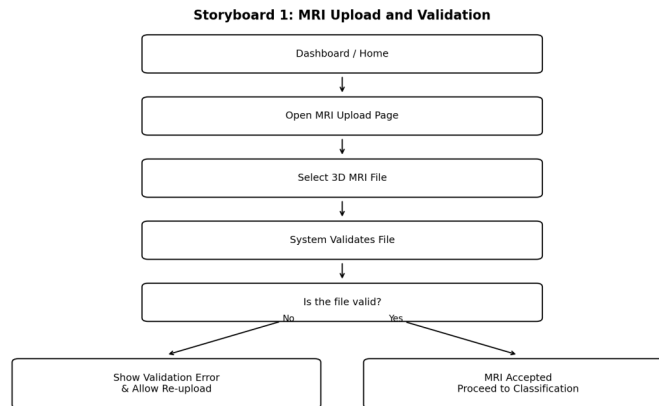


Figure E1: Storyboard for MRI Upload and Validation

E.2 Storyboard 2 - MRI Classification Process

This storyboard illustrates the main classification interaction. After a valid MRI file has been accepted, the user starts the classification process and the system displays a processing status while the proposed hybrid spatial-sequential workflow analyses the scan.

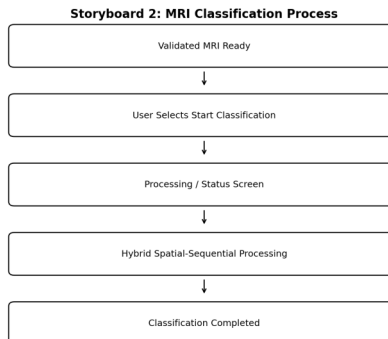


Figure E2: Storyboard for MRI Classification Process

E.3 Storyboard 3 - Classification Result Presentation

This storyboard shows how the classification output will be presented to the user. After processing is completed, the system displays the predicted classification together with confidence or other relevant supporting information in a clear result interface.

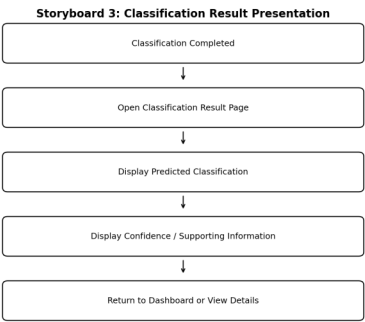


Figure E3: Storyboard for Classification Result Presentation

E.4 Storyboard 4 - Previous Classification Results

This storyboard illustrates how users can review earlier classification records. From the dashboard, the user opens the classification history, selects a previous record and views its detailed result information.

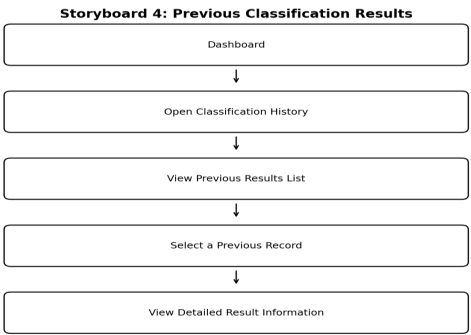


Figure E4: Storyboard for Reviewing Previous Classification Results

E.5 Storyboard 5 - Administrator Functions

This storyboard represents the planned administrative interaction. An authorised administrator can access the administrative dashboard to manage system users and stored classification records. This storyboard covers US11 and US12.

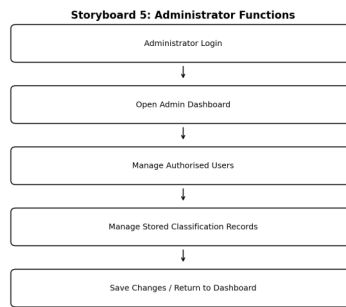


Figure E5: Storyboard for Administrator Functions

E.6 User Story Coverage

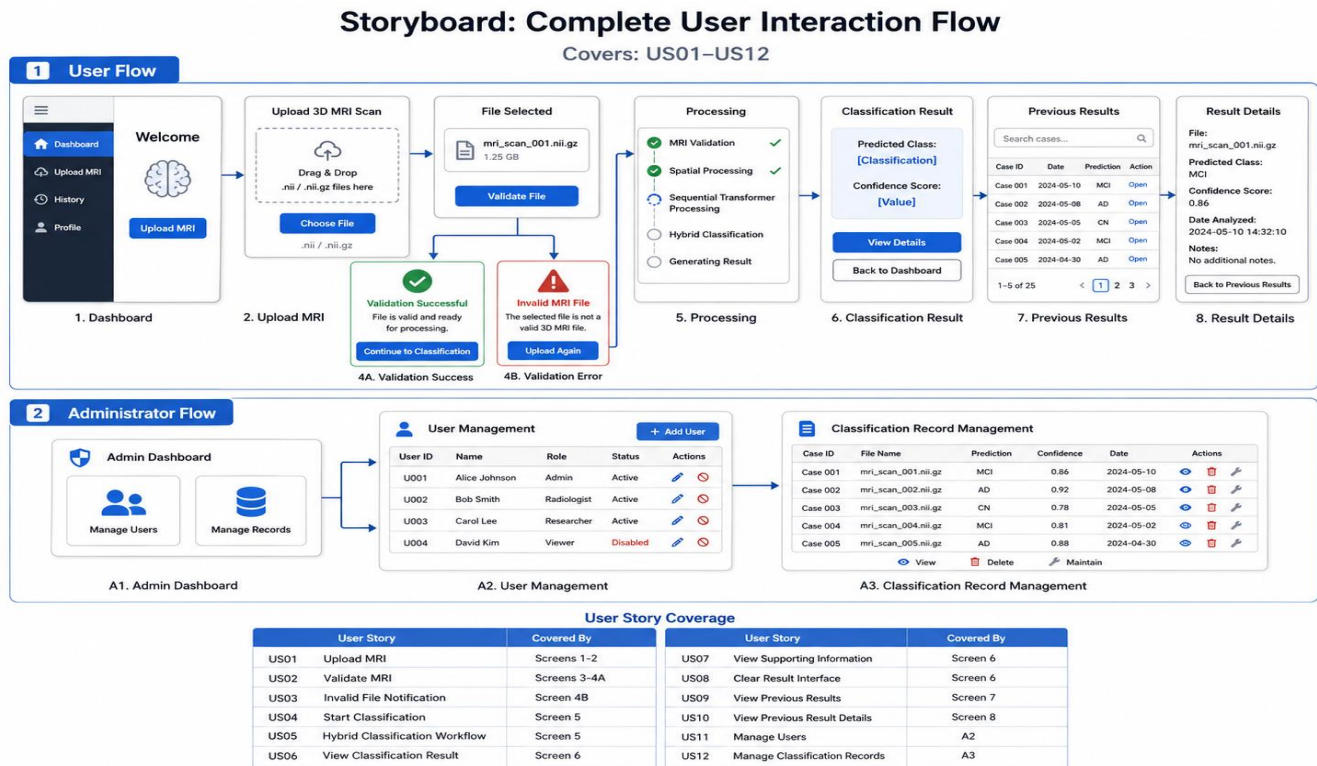


Figure:Storyboard

The five storyboards collectively cover all planned user stories and provide a visual representation of how the proposed system will be used. The interface layouts shown in the storyboards are conceptual and may be refined during implementation while preserving the planned functionality and user flows.

Conclusion

This proposal defines a structured plan for a stock market prediction system that combines machine-learning forecasting with probabilistic analysis. The intended contribution is a practical prototype that does not hide uncertainty behind a single predicted value. The project scope covers market-data acquisition and validation, feature preparation, machine-learning prediction, uncertainty quantification, time-aware evaluation, forecast visualisation, history review and administration. The WBS and Gantt plan place requirements within Planning,

complete System Design before Development, complete a Prototype before formal Testing, omit a separate Results task, and begin Documentation at the start of the project. The five planned sprints translate these activities into demonstrable increments across MIT651 and MIT652.

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